

Dylan Suvlu.

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Computational scientist working across statistical physics, high-performance molecular simulation, and machine learning.

EXPERIENCE

Sep 2020 —
Aug 2025

Postdoctoral Associate

Massachusetts Institute of Technology · Cambridge, MA

- Led molecular-dynamics and machine-learning studies of interfacial electrostatics.
- Designed and implemented GPU-accelerated Bayesian inference models that revealed previously undetected patterns in single-atom interfacial electric-field data.
- Developed high-performance, GPU-accelerated Julia implementations of Ewald summation and related electrostatics methods.
- Performed free-energy perturbation and thermodynamic-integration calculations for electron-transfer reactions at nanoparticle and electrode interfaces.
- Built end-to-end simulation and analytics pipelines that cut compute time roughly tenfold across coarse-grained polymers, polarizable nanoparticles, and interfacial electric-field systems.
- Standardized reproducible pipelines (Pixi, Docker, CI) that were adopted across projects.
- Recognized by the chemistry department for mentoring graduate and undergraduate students in research.

Sep 2014 —
Aug 2020

Graduate Student Researcher

University of Maine · Orono, ME

- Generated and analyzed ~100 TB of molecular-dynamics data to understand how water-mediated interactions govern helix-coil transitions of peptides confined in carbon nanotubes.
- Developed kinetic models of ligand-mediated nanocluster formation and growth.

EDUCATION

2014 — 2020

Ph.D., Physical Chemistry

University of Maine · Orono, ME

2010

B.S., Chemistry

University of Maine · Orono, ME

AWARDS AND FUNDING

NSF Postdoctoral Research Fellowship in Biology

National Science Foundation · 2022–2024

Postdoctoral Enrichment Program Award

Burroughs Wellcome Fund · 2021–2024

Mentorship Spotlight Award

MIT Department of Chemistry · 2022

Sequoyah Fellow

American Indian Science and Engineering Society · 2016

TECHNICAL SKILLS

LANGUAGES

Python, Julia, C++, Fortran

MACHINE LEARNING AND STATISTICS

PyTorch, PyMC, scikit-learn, XGBoost

HIGH-PERFORMANCE COMPUTING

CUDA, OpenMP, MPI, Numba, SLURM

MOLECULAR SIMULATION

OpenMM, GROMACS, LAMMPS, HOOMD-blue, RDKit, VMD, PyMOL

DATA AND INFRASTRUCTURE

NumPy, pandas, Polars, Dask, AWS (S3, Batch), Docker, Pixi, Git

PUBLICATIONS

- 2025 **Electric Fields at Solid-Liquid Interfaces: Insights from Molecular Dynamics Simulation**
J. A. Nauman, **D. Suvlu**, A. P. Willard
Annual Review of Physical Chemistry **76**, 181–202
- 2024 **Water molecules mute the dependence of the double-layer potential profile on ionic strength**
A. Limaye[†], **D. Suvlu**[†], A. P. Willard
Faraday Discussions **249**, 267–288
- 2022 **Bottlebrush polymers with flexible enantiomeric side chains display differential biological properties**
H. V.-T. Nguyen, Y. Jiang, S. Mohapatra, W. Wang, J. C. Barnes, N. J. Oldenhuis, K. K. Chen, S. Axelrod, Z. Huang, Q. Chen, M. R. Golder, K. Young, **D. Suvlu**, Y. Shen, A. P. Willard, M. J. A. Hore, R. Gómez-Bombarelli, J. A. Johnson
Nature Chemistry **14**, 85–93
- 2021 **Water-mediated interactions determine helix formation of peptides in open nanotubes**
D. Suvlu, D. Thirumalai, J. C. Rasaiah
The Journal of Physical Chemistry B **125**, 817–824
- 2020 **Nanocluster Growth and Coalescence Modulated by Ligands**
D. Suvlu[†], M. Farshad[†], J. C. Rasaiah
The Journal of Physical Chemistry C **124**, 17340–17346
- 2019 **Ligand-Mediated Nanocluster Formation with Classical and Autocatalytic Growth**
M. Farshad[†], **D. Suvlu**[†], J. C. Rasaiah
The Journal of Physical Chemistry C **123**, 29954–29963
- 2017 **Thermodynamics of Helix–Coil Transitions of Polyalanine in Open Carbon Nanotubes**
D. Suvlu, S. Samaratunga, D. Thirumalai, J. C. Rasaiah
The Journal of Physical Chemistry Letters **8**, 494–499

[†] Equal contribution.